B.Voc.(Food Processing and Quality Management)

Session 2016-17

Semester I

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-101	Communication Skills	Generic	Theory	20	80	-	-	6
*GEN-102	Fundamentals of Information Technology	Generic	Theory	20	80	-	-	6
FPQM-103	Introduction to bakery and confectionary	Skill	Theory& Practical	10	40	10	40	6
FPQM-104	Dairy technology	Skill	Theory& Practical	10	40	10	40	6
FPQM-105	Food Quality control	Skill	Theory & Practical	10	40	10	40	6

Semester II

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-201	Soft Skills and Personality Development	Generic	Theory	20	80	-	-	6
GC 202	Business Ethics	Generic	Theory	20	80	-	-	6
FPQM-203	Food Packaging	Skill	Theory & Practical	10	40	10	40	6
FPQM-204	Industrial Safety, Hazards & Prevention	Skill	Theory & Practical	10	40	10	40	6
FPQM-205	Food Plant Layout& waste disposal	Skill	Theory & Practical	10	40	10	40	6

***Refers to all Generic Components common to all B.Voc. Courses**

Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce These credits will be evaluated in Semester VI

Semester I

Code	Skill Component	Job Description
FPQM-103	Introduction to bakery and confectionary	**Baking technologist:** Responsible for manufacture of bakery and confectionary products and their quality control
FPQM- 104	Dairy Technology	**Dairy Technologist:** To supervise manufacture of dairy products like milk, butter, yogurt, cheese and their preservation
FPQM-105	Food Quality control	**Food quality control manager**: To determine and establish quality standards for food products and Is responsible for ensuring that food products produced meet standards set by both the organisation and regulatory authorities.

Semester II

Code	Skill Component	Job Role
FPQM 203	Food Packaging	**Food Packaging technician:**monitors packaging of various foodsand their subsequent compatibilityand handles all categories of packaging such as primary, secondary and tertiary packaging for food products.
FPQM 204	Industrial Safety, Hazards & Prevention	To evaluate values of industrial safety and hygiene. Is responsible for monitoring and assessing hazardous and unsafe situations and developing measures to assure workers safety
FPQM205	Food Plant Layout and waste disposal	**Food Plant layout officer:** Managing and updating food plant layout time to time as per business policies and safety standards.

Semester I

FPQM-103: INTRODUCTION TO BAKERY AND CONFECTIONERY

Course Objectives:

- To understand the roleof different ingredients used in Baking Industry.
- To know the general methods of processing and preservation of foods.
- To identify the microorganisms that can otherwise spoil bakery products.

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Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

Unit I

Introduction, Types and processing methods: Wheat: Structure, types, milling**Rice**: Structure and processing. **Flour**: grades,treatment and Storage. **Raising and Thickening Agents**: their uses .**Leavening agents:**their role in baking industry

Unit II

Methods Of Preservation: Preservation by heat treatment –Pasteurization, sterilization, blanching and canning – Heat resistance of microorganisms and their spores (bacteria, fungi and yeast). Thermal death curves, types of retorts, aseptic packaging.Spoilage of canned foods and types of spoiled cans.Cold preservation and processing. Non-thermal methods of preservation : High pressure processing

. Unit III

Introduction to Baking and confectionary process methods: Methods of Bread Making:Straight dough method, Sponge dough method, principle and types of pastry making, Types of cake making methods: Sugar batter, Flour batter method, process of biscuit making.

Unit IV

Bread Spoilage: Rope and Mould, Factors responsible for it and preventive measures.
Defects and Remedies: Basic reasons and suggested remedies: Bread, Cake and Biscuits.

Text books/ References:

1. Frasier.1987. Food Microbiology.Tata McGraw-Hill Education.
2. Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2nd Edition
3. Bread: A baker's book of techniques and recipes by Jeffrey Hamelman
4. Ornamental Confectionary and the Art of Baking in all its Branches by Herman Hueg
5. . Bakery-I, Student handbook and practical manual, published by CBSE.
6. A professional Text to bakery and confectionary by John Kingslee

Practical based on FPQM-103

Time: 3 hours

1. Preparation of White Bread.
2. . To determine moisture in bread and biscuits.
3. Studying the effect of temperature on process of biscuit making.
4. Determination of ash content in bread and biscuits.
5. Principle and preparation of Fruit cake
6. To do icing on the cake
7. . Determination of gluten

FPQM 104: DAIRY TECHNOLOGY

Course Objectives:

- To develop knowledge among students about various aspects of dairy industry.
- To study quality standards and production of various types of milk and milk products.
- To study the role of dairy farming in Indian economy

Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

Unit I

Milk: Definition, composition, physicochemical properties of milk and its constituents, nutritional value, overview of processes used in milk and milk products- pasteurization, homogenization and standardization, dye detection test, platform tests, milk borne diseases. Microbiology of milk,types of spoilage and its preventions.

Types of market milk: Toned milk, Double Toned milk, Standardized Milk, Recombined milk, reconstituted milk, fortified milk,.Effluent treatment of dairy plant disposal.

Unit II

Technology underlying manufacture ofMilk products : Butter, cream, butteroil, cheese, yoghurt, paneer, chhana, ice-cream

Unit III

Composition, standards, manufacturing: Process, equipments and defects duringmanufacturing and storage of dairy products and by products (paneer, dahi, milk powder-skimmed milk and whole milk powder, casein, whey concentrate, lactose, ghee residue).

Unit IV

Dairy development in India: - Present status, future prospective and its role inIndian economy. Important government initiative. Role of dairy development organizations (NDRI,Amul) in dairy development

Text books/ References:

1. Fluid milk industry, J.S Handerson, A.V.I Publishing Company, USA1
2. Indian Dairy products, K.T. Acharya Publication
3. Milk Hygiene in milk production processing and distribution, F.A.O Publication.
4. Sukumar.1983.Outlines of Dairy Technology.De,Oxford University Press.

PRACTICAL BASED ON FPQM 104

1. Sampling of milk and milk products for microbiological analysis
2. Platform test for milk analysis.
3. To determine fat in milk by gerber method
4. Todetermine specific gravity of given sample of milk by lactometer
5. Visit and study a nearby milk union/ dairy and prepare a checklist of problems in procurement and milk distribution
6. Detection ofvarious adulterants in milk
7. Preparation of sterilized flavored milk
8. To prepare pasteurized milk
9. Methylene blue reduction test in milk

FPQM 105: FOOD QUALITY CONTROL

Course Objectives:

• To understand the different principles and functions of food quality control department.
• To understand various food laws and regulations

Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

Unit 1

Objectives, functions and principles of quality control .Difference between food quality control and quality assurance, assessment of raw materials and finished products (quality control)of baking industry.

Unit II

Total quality management (TQM) – Principles of quality management, good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control.

Quality Attributes: Size, shape, colour, aroma and texture. Microbial quality control: an overview

Unit III

Food laws and regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO 9000 series etc, Food standards and safety Act: salient provisions and prospects, role of various food standards in india- PFA, FPO, AGMARK and BIS.

Unit IV

Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control.

Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of bakery products

Practical based on FPQM- 105

1 Techniques of quality assessment of different natural and processed foods.
2 Documentation of details of baking ingredients, process and finished products used in baking industry.
3 Quality evaluation of milk and milk products
4 Sensory methods for measuring foodquality assessment of raw materials used and processed bakery products
5 Instrumental methods for measuring foodquality assessment of raw materials used and processed bakery products
6 Study of cleaning and sanitizers used in pre and post-operative processes in bakery industry
7 Listing of quality control agencies at national and international levels

Text books/ References:

1. InteazAlli (2003). Food Quality Assurance Principles and Practices.CRC Press.

2. J Andres Vasconcellos (2003) Quality Assurance for the Food Industry. A Practical Approach.CRC Press

Semester II

Paper Title: GC 202 - BUSINESS ETHICS

Objective: The objective of this paper is to familiarize the students with the importance of ethicsin business and understanding of issues related to corporate social responsibility and corporate governance.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit 1

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics.

Unit-II

Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics.Ethical Issues in Indian Business.

Unit - III

Professional Values:Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, spirituality

Unit-IV

Corporate Social Responsibility**:** Social Responsibility of business with respect to different

stakeholders, Arguments for and against social responsibility of business, Social Audit,Corporate Social Responsibility and Corporate Governance.

Suggested Readings:

1. J.P. Sharma, *Corporate Governance, Business Ethics &CSR,*Ane Books Pvt. Ltd., Nee Delhi.
2. Andrew Crane, Dirk Matten, *Business Ethics,* Oxford University Press, New Delhi.
3. Daniel Albuquerque, *Business Ethics, Principles and Practices (Indian Edition),* Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, *Business Ethics and Social Responsibility*, Rexestore.
5. O. C. Ferrell, John Fraedrich, Linda Ferrell, *Business Ethics: Ethical Decision Making &Cases, Cengagae Learning*
6. Michael Blowfield, Alan Murray, *Corporate Responsibility – A Critical Introduction,* Oxford University Press, New Delhi.

FPQM- 203: FOOD PACKAGING

Course Objective:

• To enable the students to understand about packaging and packaging materials, compatibility of various food items with packaging materials

Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

Unit I

Packaging Technology: Definitions, Functions of packaging, Properties of packaging material in relation to their functions, package design,Tests on packaging materials, Gas and water vapor transmission rates, types of containers- primary& secondary, flexible & rigid, hermetic & non-hermetic.

Unit II

Packaging materials: Flexible films and pouches, Plastics used and their Specific applications. Rigid and Semi rigid plastic packaging methods. Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card Board packaging and their applications.

Wood: structure, types of wooden containers and their properties.
Glass : composition, properties, types and manufacture of glass containers.

Unit III

Metal Cans: properties of metals for packaging material, Types of Metallic Cans - Tin cans and Aluminum cans, steel plate and functions of its various constituents, Importance of Open top sanitary cans, Lacquering- types and applications, three piece cans and two piece cans.

Introduction to Canning operations – Can Reformer, Flanger, Seaming, Can closures, Sterilization of jars and bottles.

Unit IV

Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of Flexible plastic containers,
Aseptic packaging, shrink packaging, gas packaging, vacuum and modified atmosphere packaging –principles, applications.
Estimation of shelf life, analysis of storage requirement, accelerated storage

Text Books/ References.

1. Coles, R., Dowell, D.M., Kirwan, J. "Food PackagingTechnology".BlackWellPublishing Ltd.ISBN-084939788X, 2003.

2. NIIR Board: Food Packaging Technology Handbook National Institute of Industrial Research, New Delhi (2004)

3. PirengerO.G.andA.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH, Germany (2000).

Practical based on FPQM- 203

1. To determine thickness of paper and paper board.
2. To identify different types of packaging materials.
3. Demonstration. of measurement of cartons' dimensions as per organizational standards.
4. Demonstration of measurement of dimensions of bottle mouth, cans and their caps.
5. Demonstration of sealing processes used in food industry.
6. Demonstration of filling process.

FPQM- 204: INDUSTRIAL SAFETY, HAZARDS & PREVENTION

Course Objectives:
• To create awareness about health hazards of industrial substances.
• To evaluate the threshold value of industrial hygiene and safety.

Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

UNIT-I

Microbial Contaminants associated with food–Bacteria, viruses, fungus, molds and yeast. Factors affecting the growth of microbes in food. General Microbiological Methods of enumeration and isolation of food related microbes.

General methods of Preservation: Pasteurization,sterilization and appertization

UNIT-II

Studies of process hazards, Law Codes, Standards, Properties and functions of Chemicals, and Health hazards of industrial substances.

UNIT-III

Toxicology :Toxic materials and their properties, effect of dose and exposure time, relationship and predictive models for response, Threshold value and its definitions, , industrial sanitation and hygiene evaluation.

UNIT-IV

Propagation of fire and effect of environmental factors, ventilation, dispersion, purifying and sprinkling, safety and relief valves.

Text books/ References:

1. Crawl D.A. And Louver J.A., 2001 "Chemical Process Safety Fundamentals With Applications" Prentice Hall
2. Wentz, C.A, 2001. "Safety Health And Environmental Protection," McGraw Hill
3. Smith, B.D, 2001 "Design Of Equilibrium State Process, "McGraw Hill.

Practical based onFPQM- 204

Time: 3 hours

1. Methods of sterilization and preparation of media
2. Gram staining
3. Study of morphology of bacteria, yeast and fungi
4. Methods of pure culture techniques for bacteria
5. Enumeration and isolation of bacteria and fungi from water/milk and contaminated food
6. Demonstration of different safety aspects and maintenance of material safety data sheets followed in food industry.

FPQM-205: FOOD PLANT LAYOUT AND WASTE DISPOSAL

Course objective:

Introduction of the basic setup of a food processing industry

To make them conversant with the machinery and equipments used in different types of food industry

Instructions for paper setters

The syllabus of this paper has been divided into four units.

Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus.

The students are required to attempt one question from each All questions carry equal marks.

Unit I

Industrial plant design concepts and general design considerations for location of food plants. Application of HACCP concept, ISO, FPO & MPO requirements in food plant layout and design.

Unit II

Importance of plant layout selection of site .Selection of building material.Selection and planning of manufacturing process and service facilities Basic understanding of equipment layout and ventilation in food processing plants .Process flow charts for material movement and utility consumption in food plants.

Unit III

Plant layout and design ofmilk and milk products.Plant layout and design of beverage industry

Unit IV

Plant layout and design of bakery and biscuit industries.Miscellaneous aspects of plant layout and design like provision for waste disposal, safety arrangements etc.

Text Books/ References.

1. J M Apple, "Plant layout and Material handling", John Willey & Sons, (1977)

2. Manufacturing Facilities Design and Material Handling by Fred E. Meyers, and Matthew P. Stephens, 3rd Edition, Pearson Prentice Hall, 2000.

3. James M Moore, "Plant Layout and Design", McMillan & Co., (1959)

Practical based on FPQM-205 **Time: 3 hours**

1. Industrial visit and report making.